DRILL 05

GROUND LADDERS

NFPA 1001 Level 1 (JPR 4.3.6, 4.3.12) / WorkSafeNB Reg 91-191

ROOF LADDER DEPLOYMENT & LEG LOCK WORK POSITION

⏱ **Time:** 45–60 min | 🚒 **Apparatus:** Pumper / Ladder Rack | 🦺 **PPE:** Full Structural Turnout Gear & Gloves

👥 **Roles:** Drill Leader (Facilitator) | Safety Officer (Watcher) | Ladder Climber & Ground Heeler

🎯 OBJECTIVE & RURAL FIREGROUND VALUE

Every firefighter must be able to carry a 14-ft or 16-ft roof ladder up an extension ladder, open and lock the spring hooks, slide the ladder up a pitched roof until the hooks catch the roof ridge, and transition safely onto the roof ladder while executing a secure **leg lock** or life-safety harness tie-in. Essential for rural chimney fires and roof ventilation on steep Maritime roofs.

📦 TOOLS & APPARATUS SETUP

- 24-ft or 28-ft extension ladder raised and secured to station roofline or training prop (minimum 3–5 rungs above gutter).
- 14-ft or 16-ft roof ladder with spring-loaded folding hooks.
- Flathead axe or halligan tool for sounding roof deck.

⚠️ SAFETY WATCH-OUTS & STOP RULE

⚠️ **Fall Arrest / Ladder Safety:** The base extension ladder must be firmly heeled or tied off at the base. Climbers must maintain **3 points of contact** at all times.

⚠️ **Hook Lock Verification:** Visually and physically confirm both roof hooks have clicked and locked open before the ladder leaves the ground.

⚠️ **Ridge Embedment:** The firefighter must push the roof ladder over the ridge and pull back firmly until both hooks bite deeply into the roof ridge board.

🛑 **THE STOP RULE: Shout "STOP!" immediately if the climber loses balance, if the roof ladder fails to catch the ridge, or if footing on the roof deck slips.**

PHASE 1: GROUND PREPARATION & HOOK DEPLOYMENT (10 MINUTES) — SLOW DRY RUN

1. Remove roof ladder from apparatus.
2. Inspect spring-loaded hooks: Pull outwards and pivot 180 degrees until the locking pins snap into place.
3. Test hook locking mechanism by hand pressure; confirm hooks do not collapse.

DRILL 05: PRACTICAL EVOLUTIONS & TAILGATE DEBRIEF

Ground Ladders • Practical Application

PHASE 2: CARRYING ROOF LADDER UP GROUND LADDER (20 MINUTES) — BUDDY COACHING

1. **The Shoulder Carry Method:**
   - Firefighter faces the extension ladder with roof ladder resting on their shoulder, hooks pointing forward and curled outward/downward.
   - Grasp the rungs of the extension ladder with both hands while sliding the roof ladder along on the shoulder.
   - Climber climbs smoothly, taking one rung at a time.
2. **Transition at the Eaves:**
   - Stop when your shoulders are level with the roof eaves.
   - Rest the butt of the roof ladder onto the roof deck or gutter edge.
   - Push the roof ladder upward with both hands along the roof surface, allowing the wheels (if equipped) or hooks to slide smoothly up to the peak.
   - As the hooks clear the ridge peak, rotate or pull the ladder downward firmly until both hooks **bite into the opposite side of the roof ridge**.
   - Pull hard 2–3 times to verify total engagement before placing any body weight on it.

PHASE 3: TRANSITION & LEG LOCK WORK POSITION (20 MINUTES) — TIMED / STRESS

1. **Transitioning onto Roof Ladder:**
   - Step carefully from the extension ladder onto the center rungs of the roof ladder (never step directly onto un-sounded roof deck).
   - Keep your weight centered over the beams.
2. **Executing the Leg Lock (Hands-Free Work Position):**
   - Reach your leg through the ladder rungs (one rung above your standing foot).
   - Wrap your lower leg around the beam and hook your foot behind the rung below.
   - Straighten your other leg to lock your shin and instep securely against the ladder.
   - Step back with your hands to demonstrate a secure, hands-free position (simulating operating an axe, TIC, or chimney drop-chain).
3. If using ladder safety belts, connect the carabiner/hook to the center of the rung at waist height.

⚠️ COMMON PITFALLS & FIREGROUND HAZARDS

- ❌ *Hooks not locked before ascending:* Trying to deploy hooks with gloved hands while dangling from a 20-foot ladder is dangerous and slow. Lock them on the ground!
- ❌ *Stepping between rungs on the roof deck:* Rotted or fire-weakened roof sheathing will collapse under your boots. Always step strictly on the ladder rungs.
- ❌ *Pushing ladder without looking at the ridge:* If you miss the ridge or hit a chimney flashing, the ladder can slide down and kick you off the extension ladder.

🗣️ 5-MINUTE TAILGATE DEBRIEF

1. ***"Who felt unstable while transferring from the extension ladder to the roof ladder?", "Did the hooks bite solid into the peak?"***

✅ APPARATUS RESET CHECKLIST

- ☐ Roof ladder hooks released and folded flat against beams.
- ☐ Spring-loaded pins lubricated if stiff.
- ☐ Extension ladder lowered and both ladders returned to brackets.
- ☐ Safety straps latched.
- ☐ Attendance roster signed.

DRILL LEADER SIGNATURE	SAFETY OFFICER SIGNATURE	DATE COMPLETED	TRAINING HOURS